



LAIBA MUGHAL

70077

BS SE – 3

Data Structures Lab

Lab NO# 14

Lab Task 1:

1. Undirected Graph: A-B-C-D-E-F

Edges: A-B, B-C, B-D, C-E, D-E, E-F

Adjacency List

- A: B
- B: A, C, D
- C: B, E
- D: B, E
- E: C, D, F
- F: E

Adjacency Matrix - Order: A B C D E F

	A	B	C	D	E	F
A	0	1	0	0	0	0
B	1	0	1	1	0	0
C	0	1	0	0	1	0
D	0	1	0	0	1	0
E	0	0	1	1	0	1
F	0	0	0	0	1	0

BFS from A:

$A \rightarrow B \rightarrow C, D \rightarrow E \rightarrow F$

DFS from A:

$A \rightarrow B \rightarrow C \rightarrow E \rightarrow D \rightarrow F$

2. Directed Graph:

$A \rightarrow D, A \rightarrow E, B \rightarrow A, C \rightarrow B, C \rightarrow E, E \rightarrow F$

Adjacency List

- A: D, E
- B: A
- C: B, E
- D:
- E: F
- F:

BFS from A:

$A \rightarrow D, E \rightarrow F$

DFS from A:

$A \rightarrow D, A \rightarrow E \rightarrow F$

3. Weighted Graph:

A-B-6, A-C-6, A-D-8, B-E-7, C-E-6, E-F-8

Adjacency List with weights

- A: (B,6), (C,6), (D,8)
- B: (A,6), (E,7)
- C: (A,6), (E,6)
- D: (A,8)
- E: (B,7), (C,6), (F,8)
- F: (E,8)

BFS from A:

$A \rightarrow B, C, D \rightarrow E \rightarrow F$

DFS from A:

$A \rightarrow B \rightarrow E \rightarrow C \rightarrow F \rightarrow D$

Lab Task 2: Airline routes

Cities: Boston=B, Newark=N, Miami=M, Detroit=D, Washington=W

Type of graph:

Depends on the part:

1. Simple graph:

1 edge if any flight exists between cities.

Edges:

B-N, N-M, N-D, N-W, W-M

2. Multigraph: 1 edge per flight.

Edges:

4x B-N, 2x N-B, 3x N-M, 2x M-N, 1x N-D, 2x D-N, 3x N-W, 2x W-N, 1x W-M

3. Multigraph + loop:

Same as #2 + loop at M for sightseeing trip

Lab Task 3:

Class name-knowing graph

Model:

Use a directed graph. Each person = vertex.

- Directed?

Yes. Person X knowing Y's name doesn't mean Y knows X's name.

- Multiple edges?

No. We either know the name or we don't. Multiple edges don't make sense here.

- Loops?

Yes, if we assume everyone knows their own name. No, if we only care about knowing others. Most cases: no loops.

Lab Task 4: The two numbered/lettered graphs

Left graph with nodes 0-8:

I'll assume start = 0

BFS from 0:

$0 \rightarrow 1, 3, 8 \rightarrow 7, 2, 4 \rightarrow 6, 5$

DFS from 0:

$0 \rightarrow 1 \rightarrow 7 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 8 \rightarrow 5 \rightarrow 6$

Right graph with A-H, S:

I'll assume start = S

BFS from S:

$S \rightarrow A, C, G \rightarrow B, F, H \rightarrow D, E$

DFS from S:

$S \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H$